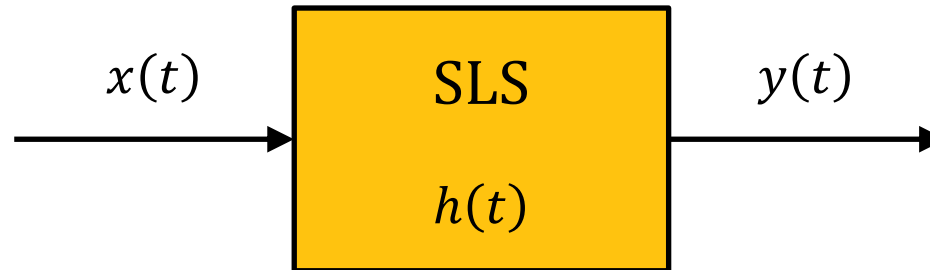


Transfer Function and Stability of Stationary Linear Systems

TF and Stability of SLSs

SYSTEM, TRANSFER FUNCTION



$$y(t) = h(t) * x(t) = \int_0^t h(\tau)x(t - \tau)d\tau$$



Borel's convolution theorem

$$Y(s) = H(s)X(s)$$

SLS means Stationary Linear System, for which if:

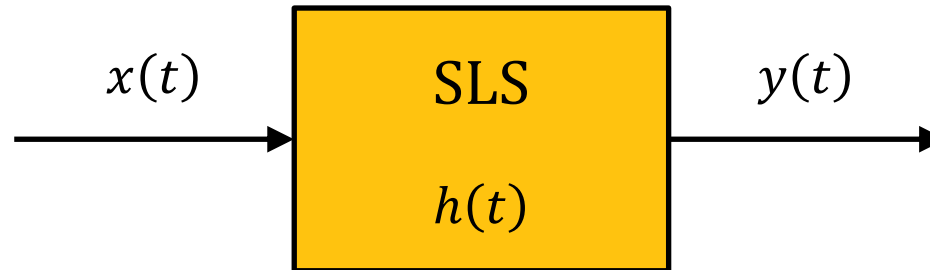
$$x(t) \leftrightarrow y(t)$$

then:

$$ax(t - \tau) \leftrightarrow ay(t - \tau)$$

TF and Stability of SLSs

SYSTEM, TRANSFER FUNCTION



The function $H(s)$ that relates reaction $Y(s)$ to excitation $X(s)$ is called a **transfer function of the system**. Its time-domain form $h(t)$ is called **impulse response**:

If the excitation has a form of Dirac's delta:

$$x(t) = \delta(t)$$

then its Laplace transform is equal to unity:

$$X(s) = 1$$

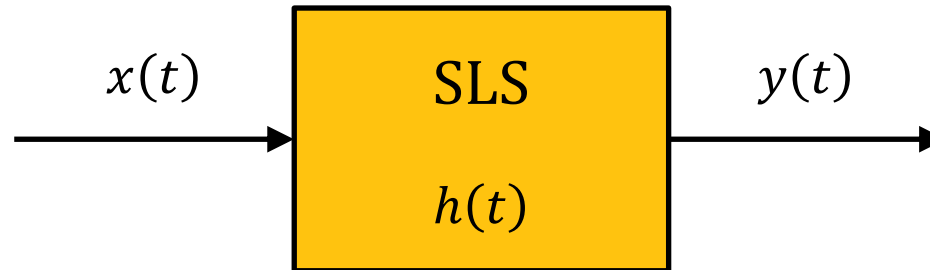
For such a case:

$$Y(s) = H(s)X(s) = H(s)$$

Hence the system's response $y(t)$ is equal to the system's transfer function $h(t)$.

TF and Stability of SLSs

SYSTEM, TRANSFER FUNCTION



The transfer function is a rational function i.e., it is a ratio of two polynomials of variable s with real coefficients:

$$H(s) = \int_{0-}^{\infty} h(t)e^{-st} dt = \frac{L(s)}{M(s)}$$

where

$$L(s) = \sum_{i=0}^N a_i s^i, \quad M(s) = \sum_{i=0}^N b_i s^i, \quad a \in \mathbb{R}, \quad b \in \mathbb{R}$$

Transfer function can be expressed as ratio of voltages $\frac{U_{OUT}(s)}{U_{IN}(s)}$, currents $\frac{I_{OUT}(s)}{I_{IN}(s)}$, or mixed $\frac{U_{OUT}(s)}{I_{IN}(s)}, \frac{I_{OUT}(s)}{U_{IN}(s)}$.

TF and Stability of SLSs

BIBO STABILITY

The system is stable in the sense of BIBO (Bounded Input Bounded Output) when the relationship is fulfilled:

$$|x(t)| < \infty \text{ for } t \geq t_0 \Rightarrow |y(t)| < \infty \text{ for } t \geq t_0, t_0 \in \mathbb{R}$$

Bounded excitation triggers a bounded response

What properties must the system have to exhibit bounded response in BIBO sense?

TF and Stability of SLSs

BIBO STABILITY

What properties must the system have to exhibit bounded response in BIBO sense?

System's impulse response has a general form of:

$$h(t) = a\delta(t) + h_0(t)$$

The system exhibits bounded response to a bounded excitation if:

$$\int_0^{\infty} |h(t)| dt = |\alpha| + \int_0^{\infty} |h_0(t)| dt < \infty$$

The above approach is, however impractical. By using Laplace transform the BIBO stability can be verified easier.

TF and Stability of SLSs

BIBO STABILITY

Since the transfer function $H(s)$ is a rational function, it can be decomposed into partial fractions:

$$H(s) = \frac{L(s)}{M(s)} = \frac{L(s)}{\prod_{i=1}^N (s - s_i)} = \sum_{i=1}^N \frac{L_i(s)}{s - s_i}$$

where:

$$\text{ord}[L(s)] \leq \text{ord}[M(s)]$$

and s_i are **poles** of the transfer function i.e.:

$$s = s_i \leftrightarrow M(s) = 0.$$

On the other hand, values s_k for which:

$$s = s_k \leftrightarrow L(s) = 0$$

are called **zeroes** of the transfer function.

TF and Stability of SLSs

BIBO STABILITY

Using the Laplace transform table one can notice, that:

$$H(s) = \sum_{i=1}^N \frac{L_i(s)}{\boxed{s - s_i}} \qquad h(t) = \sum_{i=1}^N f_i(t) \boxed{e^{s_i t}}$$


Hence, if all poles of the transfer function are in left half-plane i.e., for each i :

$$\Re[s_i] = \sigma_i < 0$$

and:

$$\text{ord}[L(s)] \leq \text{ord}[M(s)]$$

then the system described by $H(s)$ is stable in BIBO sense.

TF and Stability of SLSs

BIBO STABILITY – HURWITZ POLYNOMIAL

Denominator of the transfer function is equal to:

$$M(s) = \prod_{i=1}^N (s - s_i) = \sum_{i=1}^N a_i s^i, \text{ where } a_i \in \mathbb{R}$$

A polynomial having roots in left half-plane only is called **Hurwitz polynomial**.

Hence, then the system described by:

$$H(s) = \frac{L(s)}{M(s)}$$

is stable in BIBO sense if:

1. $\text{ord}[L(s)] \leq \text{ord}[M(s)]$
2. $M(s)$ is a Hurwitz polynomial

(both conditions must be met)

TF and Stability of SLSs

HURWITZ POLYNOMIAL ANALYSIS

There is a number of methods that can provide information on system stability without calculating the elements of the equation

$$M(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0 = 0$$

This equation can also be written as:

$$M(s) = a_n (s - s_n)(s - s_{n-1}) \dots (s - s_0)$$

Example – 1st order

$$n = 1$$

$$a_1 > 0$$

$$M(s) = a_1 s + a_0 = a_1 (s - s_1) \rightarrow s_1 = -\frac{a_0}{a_1}$$

1. $s_1 < 0 \rightarrow a_0 > 0$ (when $a_0 > 0$),
2. $a_0 > 0 \rightarrow s_1 < 0$ (when $a_0 > 0$),

The necessary and sufficient condition for $M(s)$ of order $n = 1$ to be a Hurwitz polynomial (HP) is that $a_0 > 0$ and $a_1 > 0$.

TF and Stability of SLSs

HURWITZ POLYNOMIAL ANALYSIS

Example – 2nd order

$n = 2$ $a_2 > 0$

$$M(s) = a_2 s^2 + a_1 s + a_0 = a_2 (s - s_2)(s - s_1)$$

From Viète formulas:

$$\frac{a_1}{a_2} = -(s_2 + s_1), \quad \frac{a_0}{a_1} = s_1 s_2$$

For a BIBO stable system two cases may occur:

1. $s_1, s_0 \in \mathbb{R}, s_1 < 0$ and $s_0 < 0 \rightarrow a_1 > 0$ and $a_0 > 0$,

Hence $a_1 > 0$ and $a_0 > 0$

2. $s_1, s_0 \in \mathbb{C}, s_1 = s_0^*, \Re[s_1] = \Re[s_0] < 0$

Hence $a_1 > 0$ and $a_0 = s_1 s_0 = |s_1|^2 > 0$

The necessary and sufficient condition for $M(s)$ of order $n = 2$ to be a Hurwitz polynomial (HP) is that $a_1 > 0$ and $a_0 > 0$

TF and Stability of SLSs

HURWITZ POLYNOMIAL ANALYSIS

Polynomials of higher orders

$$\begin{array}{l} n > 2 \\ a_n > 0 \end{array}$$

It can be shown that the necessary (but not sufficient) condition for a polynomial of the n^{th} order

$$M(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0$$

to be a Hurwitz polynomial (HP) is that the polynomial coefficients meet the following conditions:

$$a_n > 0, a_{n-1} > 0, a_{n-2} > 0, \dots, a_0 > 0$$

Only for polynomials of the first and second order $M(s)$, this condition also becomes sufficient.

For example, it can be shown that for a polynomial $M(s)$ of the third order:

$$M(s) = a_3 s^3 + a_2 s^2 + a_1 s + a_0$$

additional, sufficient condition $a_2 a_1 > a_3 a_0$ must be met for $M(s) \in HP$

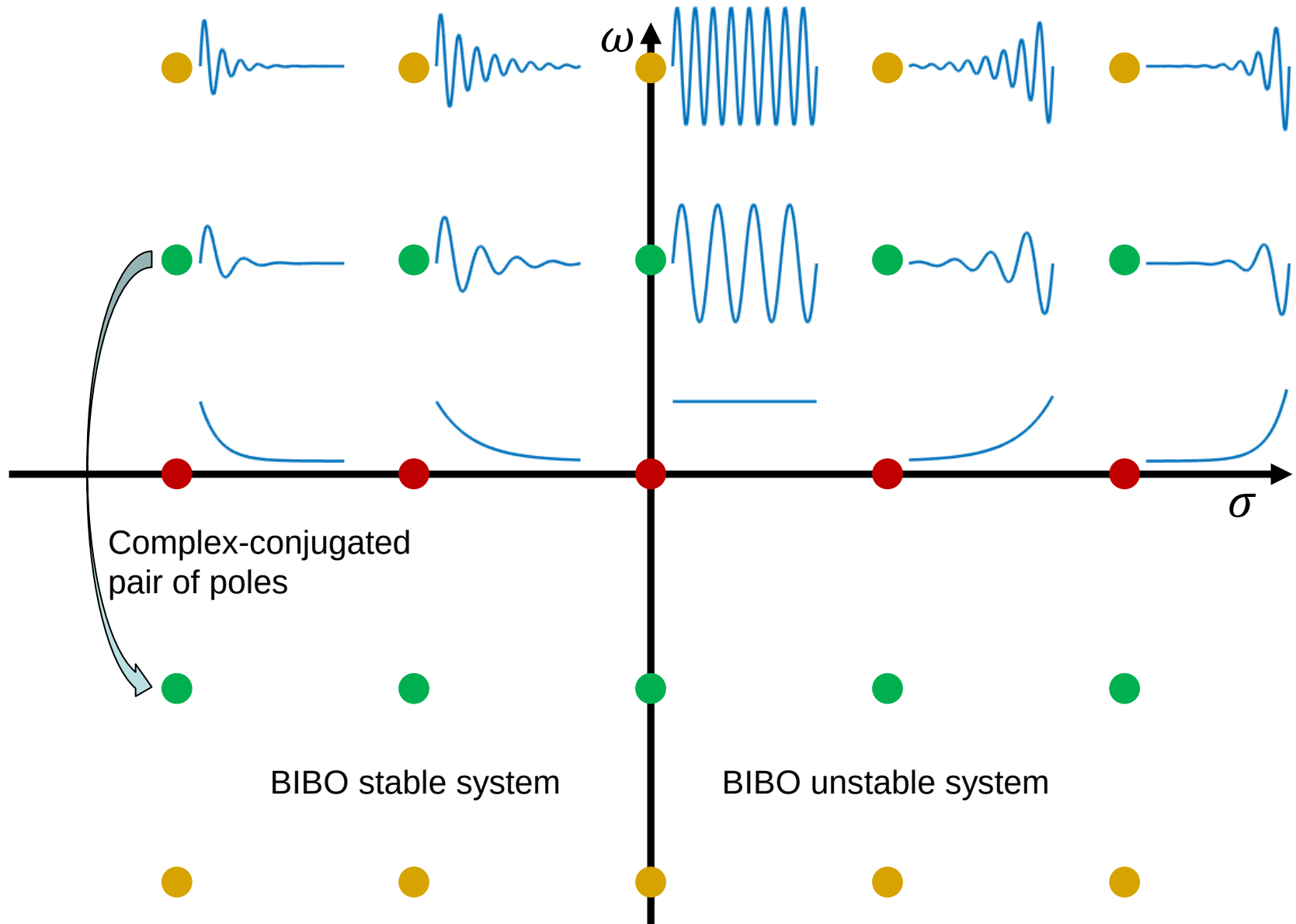
Example

$$M(s) = 3s^3 + s^2 + 2s + 8, \quad 2 \cdot 1 > 3 \cdot 8 \rightarrow M(s) \notin HP$$

TF and Stability of SLSs

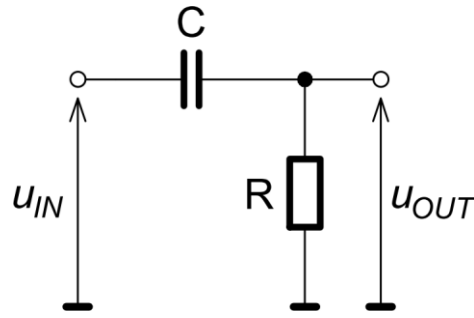
TF POLES vs. OUTPUT RESPONSE

$$s = \sigma + j\omega$$



TF and Stability of SLSs

EXAMPLE 1



$$H(s) = \frac{U_{OUT}}{U_{IN}} = \frac{R}{R + \frac{1}{sC}} = \frac{sRC}{sRC + 1}$$

$$RC > 0 \rightarrow M(s) = sRC + 1 \in HP$$

Circuit is BIBO stable

$$R = 1\Omega, C = 1F$$

$$U_{IN}(t) = \mathbf{1}(t)$$

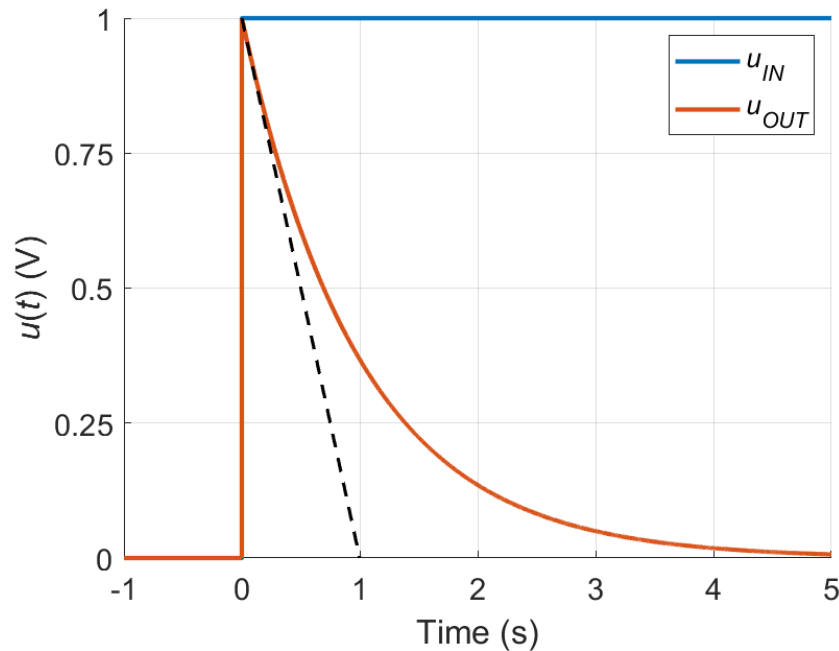
$$U_{IN}(s) = \frac{1}{s}$$

$$U_{OUT}(s) = H(s)X(s)$$

$$U_{OUT}(s) = \frac{1}{s} \frac{sRC}{sRC + 1}$$

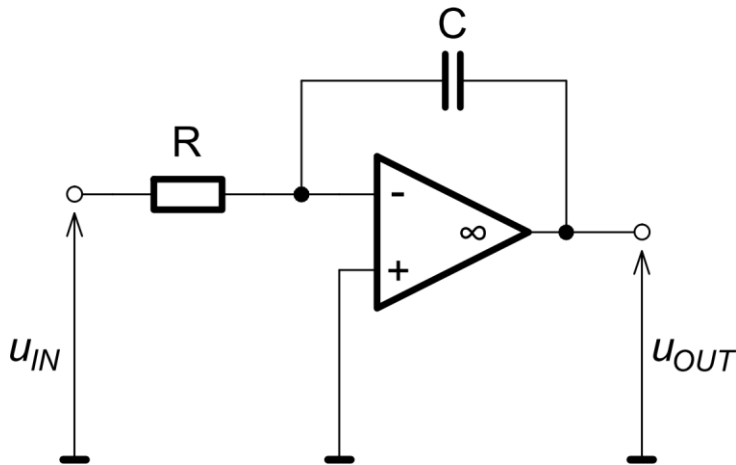
$$U_{OUT}(s) = \frac{RC}{sRC + 1}$$

$$u_{OUT}(t) = \mathbf{1}(t)e^{-\frac{t}{RC}}$$



TF and Stability of SLSs

EXAMPLE 2



$$R = 1\Omega, C = 1F$$

$$U_{IN}(t) = \mathbf{1}(t)$$

$$U_{IN}(s) = \frac{1}{s}$$

$$U_{OUT}(s) = H(s)X(s)$$

$$U_{OUT}(s) = -\frac{1}{s} \frac{1}{sRC}$$

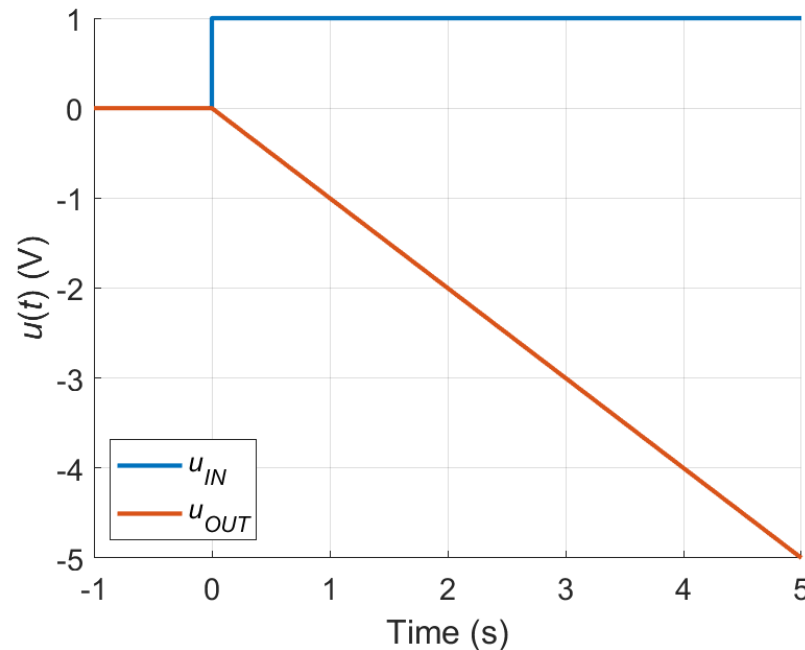
$$u_{OUT}(t) = -\frac{1}{RC} \int_0^t u_{IN}(\tau) d\tau$$

$$u_{OUT}(t) = -\mathbf{1}(t)t$$

$$H(s) = \frac{U_{OUT}}{U_{IN}} = -\frac{\frac{1}{sC}}{R} = -\frac{1}{sRC}$$

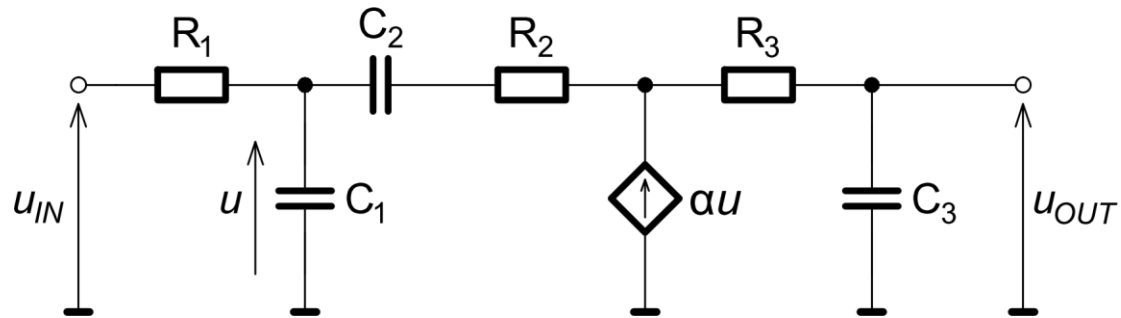
$$M(s) = sRC \notin HP$$

Circuit is BIBO unstable



TF and Stability of SLSs

EXAMPLE 3



$$R_1 = R_2 = 1\Omega, \quad R_3 = 2\Omega, \quad C_1 = C_2 = C_3 = 1F$$

$$H(s) = \frac{\alpha(s+1)}{(s^2 + (3-\alpha)s + 1)(2s+1)}$$

Circuit is BIBO stable for $\alpha < 3$

TF and Stability of SLSs

EXAMPLE 3 – CASE A

$$H(s) = \frac{\alpha(s+1)}{(s^2 + (3-\alpha)s + 1)(2s+1)}$$

$\alpha = 1$ – circuit is BIBO stable

$$U_{IN}(t) = \mathbf{1}(t)$$

$$U_{IN}(s) = \frac{1}{s}$$

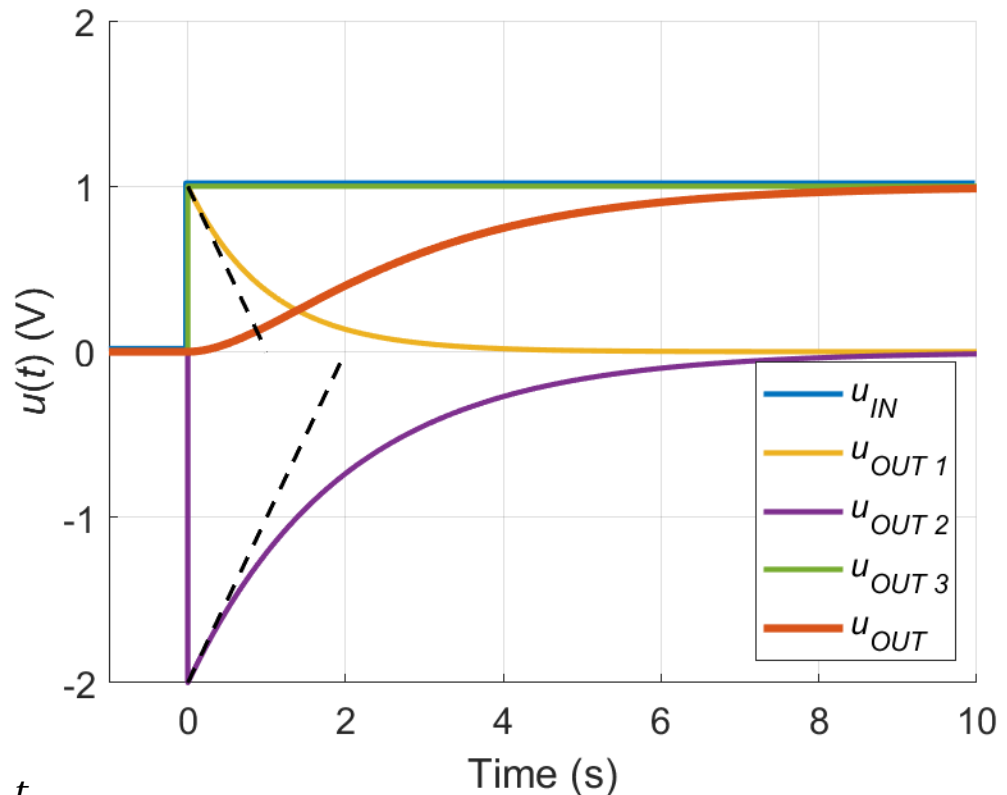
$$H(s) = \frac{1}{(s+1)(2s+1)}$$

$$U_{OUT}(s) = H(s)X(s)$$

$$U_{OUT}(s) = \frac{1}{s(s+1)(2s+1)}$$

$$U_{OUT}(s) = \frac{1}{s+1} - \frac{2}{s+\frac{1}{2}} + \frac{1}{s}$$

$$u_{OUT}(t) = \mathbf{1}(t)e^{-t} - 2 \cdot \mathbf{1}(t)e^{-\frac{t}{2}} + \mathbf{1}(t)$$



TF and Stability of SLSSs

EXAMPLE 3 – CASE B

$$H(s) = \frac{\alpha(s+1)}{(s^2 + (3-\alpha)s + 1)(2s+1)} \quad \alpha = 3 - \text{circuit is BIBO conditionally stable}$$

$$U_{IN}(t) = \mathbf{1}(t)$$

$$U_{IN}(s) = \frac{1}{s}$$

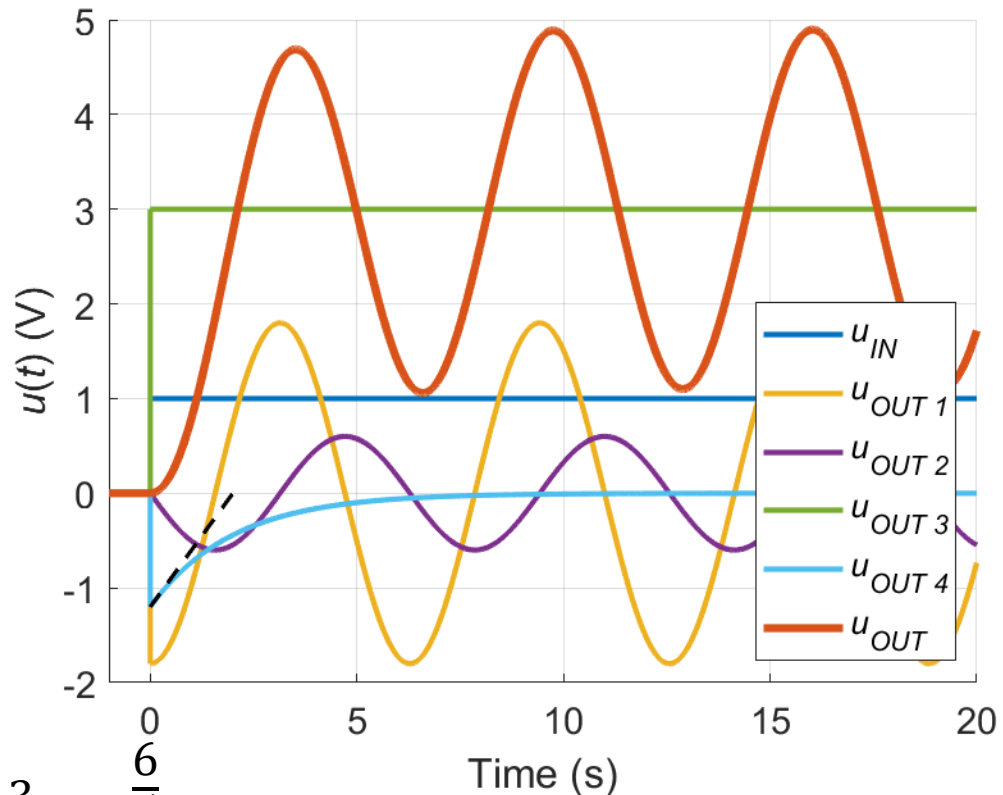
$$H(s) = \frac{3(s+1)}{(s^2 + 1)(2s+1)}$$

$$U_{OUT}(s) = H(s)X(s)$$

$$U_{OUT}(s) = \frac{3(s+1)}{s(s^2 + 1)(2s+1)}$$

$$U_{OUT}(s) = -\frac{\frac{9}{5}s}{s^2 + 1} - \frac{\frac{3}{5}}{s^2 + 1} + \frac{3}{s} - \frac{\frac{6}{5}}{s + \frac{1}{2}}$$

$$u_{OUT}(t) = -\frac{9}{5} \cdot \mathbf{1}(t) \cos t - \frac{3}{5} \cdot \mathbf{1}(t) \sin t + 3 \cdot \mathbf{1}(t) - \frac{6}{5} \cdot \mathbf{1}(t) e^{-\frac{t}{2}}$$



TF and Stability of SLSs

EXAMPLE 3 – CASE C

$$H(s) = \frac{\alpha(s+1)}{(s^2 + (3-\alpha)s + 1)(2s+1)} \quad \alpha = 5 - \text{circuit is BIBO unstable}$$

$$U_{IN}(t) = \mathbf{1}(t)$$

$$U_{IN}(s) = \frac{1}{s}$$

$$H(s) = \frac{5(s+1)}{(s-1)^2(2s+1)}$$

$$U_{OUT}(s) = H(s)X(s)$$

$$U_{OUT}(s) = \frac{5(s+1)}{s(s-1)^2(2s+1)}$$

$$U_{OUT}(s) = \frac{5}{s} - \frac{\frac{10}{9}}{s + \frac{1}{2}} - \frac{\frac{35}{9}}{s-1} + \frac{\frac{10}{3}}{(s-1)^2} +$$

$$u_{OUT}(t) = 5 \cdot \mathbf{1}(t) - \frac{10}{9} \cdot \mathbf{1}(t)e^{-\frac{t}{2}} - \frac{35}{9} \cdot \mathbf{1}(t)e^t + \frac{10}{3} \cdot \mathbf{1}(t)te^t$$

